Reg. No [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] **[14MA7615]**

SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER-2018

APPLIED MATHEMATICS-I

Time: 3 Hours **Maximum Marks:70**

Instructions:
- **i.** Answer any **FIVE** full Questions.
- **ii.** Missing data may be suitably assumed.

1. a. Find the correlation coefficient $\rho(X,Y)$ for the following joint distribution. **05**

Y \ X	-2	-1	4	5
1	0.1	0.2	0	0.3
2	0.2	0.1	0.1	0

b. Find the fixed probability vector for the matrix $A = \begin{pmatrix} 0 & 1 & 0 \\ 1/6 & 1/2 & 1/3 \\ 0 & 2/3 & 1/3 \end{pmatrix}$. **05**

c. A student's study habit is as follows: If he studies one night, he is 70% sure not to study the next night. On the other hand if he does not study one night, he is 60% sure not to study the next night. In long run how often does he study? **04**

2. a. X and Y are independent random variables. X takes the values 2, 5, 7 with probabilities 1/2, 1/4, 1/4 respectively. Y takes the values 3, 4, 5 with probabilities 1/3, 1/3, 1/3. **05**

- **i.** Find the joint distribution of X and Y.
- **ii.** Show that covariance of X and Y is zero.

b. If X and Y are continuous random variables having the joint density function **05**

$$f(x) = \begin{cases} c(x^2+y^2), 0<x<1, 0<y<1 \\ 0, otherwise \end{cases}$$

Find:

- **i.** c
- **ii.** p(x<1/2, y>1/2)

c. Gamblers luck follows a pattern. If he wins a game, the probability of winning the next game is 0.6. However if he loses a game, the probability of losing the next game is 0.7. There is an even chance of gambler winning the first game. If so in long run, how often he will win? **04**

3. a. Calculate the Fisher's ideal index from the following data and prove that it satisfies both the time reversal and factor reversal tests. **05**

Commodities	Base year		Current year	
	Price	Quantity	Price	Quantity
A	6	10	6.4	12
B	4.5	20	5.5	24
C	2.30	15	4	16
D	6.1	6	8	10

b. Using the method of least squares fit a straight line of the form p = aw+b from the following data: **05**

w	50	70	100	120
p	12	15	21	25

Find p when w is 150.

c. From the fixed base index numbers given below prepare chain base index numbers. **04**

Year	1968	1969	1970	1971	1972	1973
Index Number	376	392	408	380	392	400

4. a. Obtain the reduction formula for $\int_0^{\pi/2} \sin^n x\,dx$. Where 'n' is positive integer. **05**

b. Evaluate $\int_0^1 x^2(1-x^2)^{3/2}\,dx$. **05**

c. Evaluate $\int_0^a x\sqrt{ax-x^2}\,dx$. **04**

5. a. Trace the curve $r = a(1+\cos\theta)$. **05**

b. Find the area of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ by double integration. **05**

c. For the curve $r = ae^{\theta\cot\alpha}$,where a and α are constants, prove that **04**

$$\frac{ds}{dr} = \sec\alpha \quad and \quad \frac{ds}{d\theta} = r\,\mathrm{cosec}\,\alpha$$

6. a. Evaluate $\int_0^a\int_0^x\int_0^{x+y} e^{x+y+z}\,dz\,dy\,dx$. **05**

b. Evaluate $\iint_R xy\,dx\,dy$ where R is the region over positive quadrant of a circle $x^2+y^2=a^2$. **05**

c. Evaluate $\int_0^1 \int_x^{\sqrt{x}} xy\,dx\,dy$ by changing the order of integration. **04**

7. a. Derive the relation between the Beta and the Gamma functions as- $\beta(m,n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$. **05**

b. By using differentiation under integral sign evaluate $\int_0^1 \frac{e^{-x}\sin ax}{x}\,dx$. **05**

c. Evaluate $\int_0^{\pi/2} \sqrt{\cot\theta}\,d\theta$ by expressing in terms of gamma functions. **04**

8. a. Verify Green's theorem for $\int_C \left[(3x^2 - 8y^2)dx + (4y - 6xy)dy\right]$, where C is a triangle formed by a lines $x = 0,\ y = 0,\ x + y = 1$. **07**

b. Using stokes theorem, evaluate $\oint \vec{F}.dr$ *where* $\vec{F} = (x^2 + y^2)i - 2xyj$ and c is a rectangle in XY plane bounded by $x = 0,\ x = a\ \ y = 0,\ y = b$. **07**